

Loaded expression and metadata

(43363, 1021) (1021, 45)

After filtering: 1021 samples, 43363 genes

- ◆ Applied log1p transform (raw counts detected)

Selected top 5000 variable genes: (5000, 1021)

AUC for Exposure classification (Naive vs DENV): 0.948

Cluster 0: AUC = 0.959

Cluster 1: AUC = 0.999

Cluster 2: AUC = 0.990

Top 10 genes → AUC = 0.964

Top 100 genes → AUC = 0.986

Top 1000 genes → AUC = 0.990

Top 5000 genes → AUC = 0.948

Random Forest analysis complete.

Exposure AUC: 0.948

Cluster prediction AUCs:

Cluster	AUC
0	0.959082
1	0.998897
2	0.990006

Table: Exposure and Cluster AUCs

The Random Forest model achieved an AUC of 0.948 for distinguishing Naive vs. DENV exposures, indicating strong classification performance.

Cluster-level prediction was also highly accurate, with AUC values of 0.959, 0.999, and 0.990 across the three clusters.

These results show that both biological exposure status and unsupervised clusters derived from gene expression data are well-separated by the model.

Such high AUCs suggest that gene expression patterns contain distinct signatures for each condition.

AUC by gene subset:

	NumGenes	AUC
0	10	0.964443
1	100	0.985768
2	1000	0.989745
3	5000	0.948294

Table: AUC by Gene Subset

The model's performance increased as more variable genes were included, peaking around 1000 genes with an AUC of 0.990.

Performance was already strong with as few as 10 genes (AUC = 0.964), suggesting that a small subset of features carries substantial biological signal.

However, including too many genes (5000) slightly reduced AUC, possibly due to noise and overfitting.

This trend highlights the importance of selecting an optimal number of informative genes rather than using all available features.

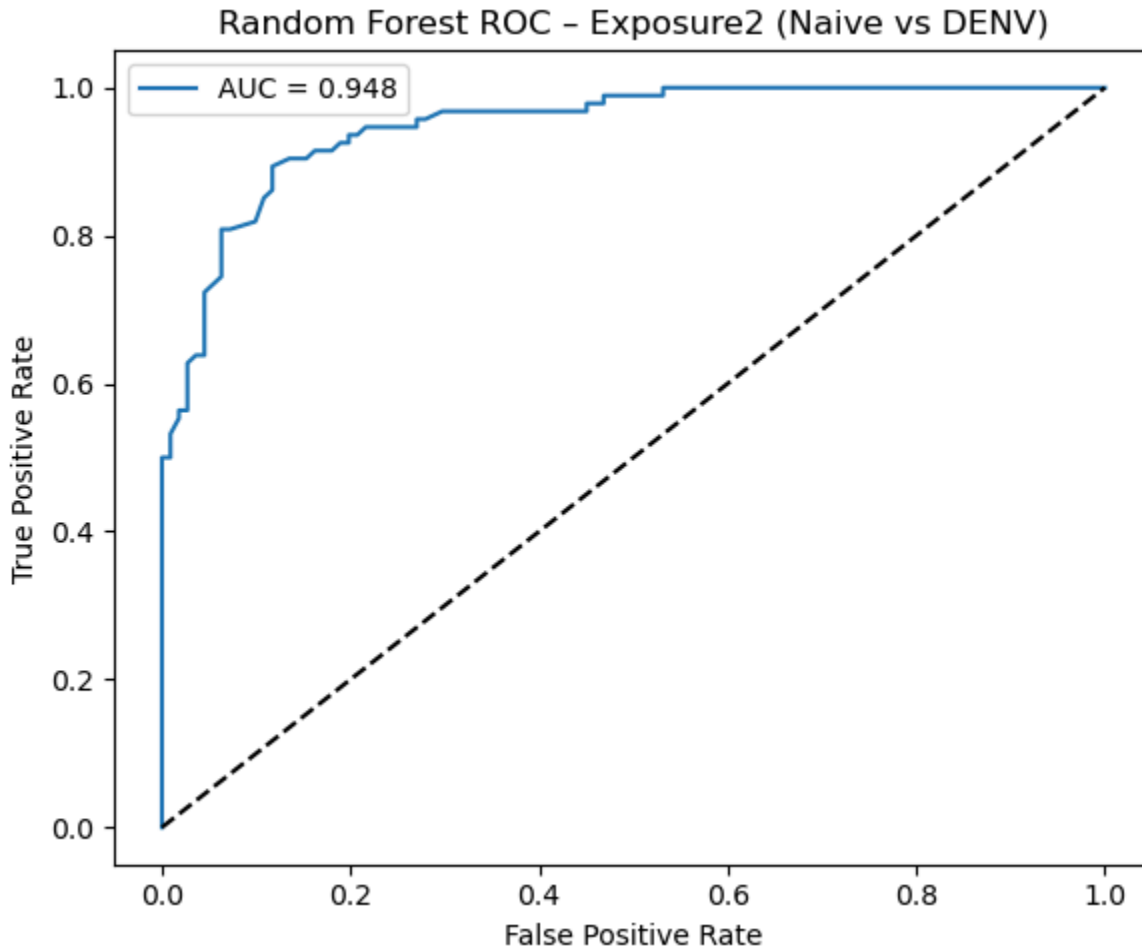


Figure 1: ROC Curve – Exposure Classification

The ROC curve shows the Random Forest model's ability to distinguish between Naive and DENV samples.

The curve rises steeply toward the top-left corner, corresponding to an AUC = 0.948, which indicates excellent classification accuracy.

The near-perfect shape suggests that gene expression differences between the two exposure groups are robust and biologically meaningful.

This confirms that the model effectively captures key transcriptional patterns linked to dengue virus exposure.

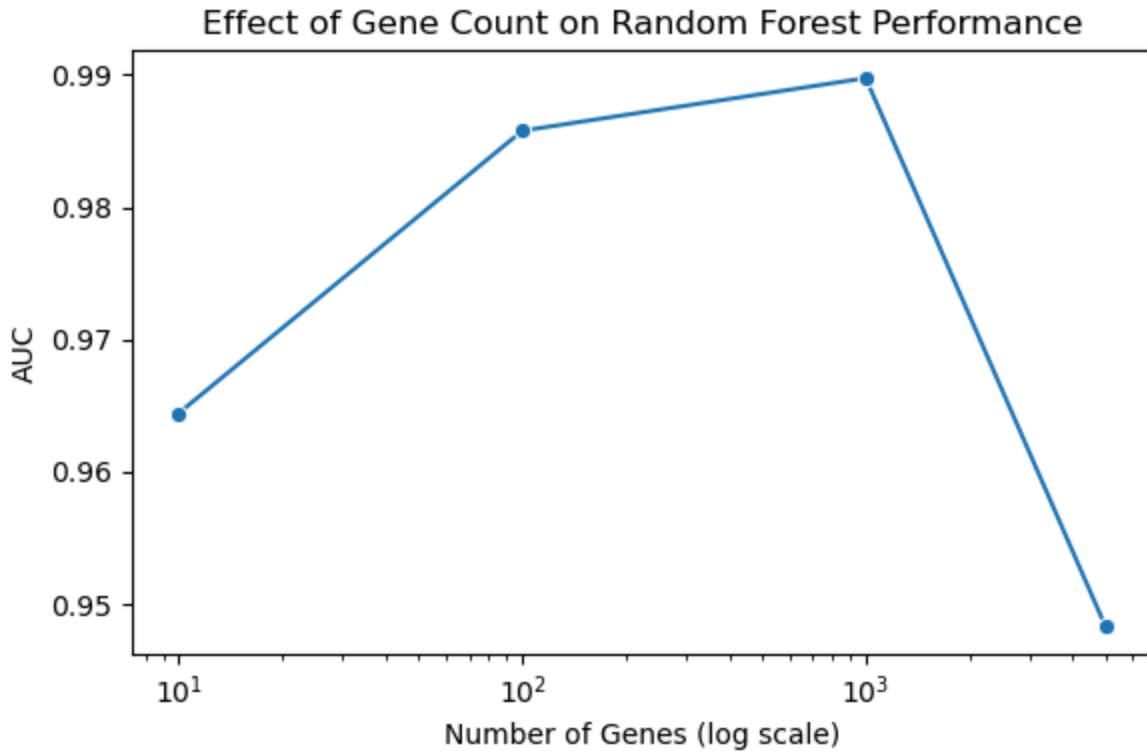


Figure 2: Effect of Gene Count on Random Forest Performance

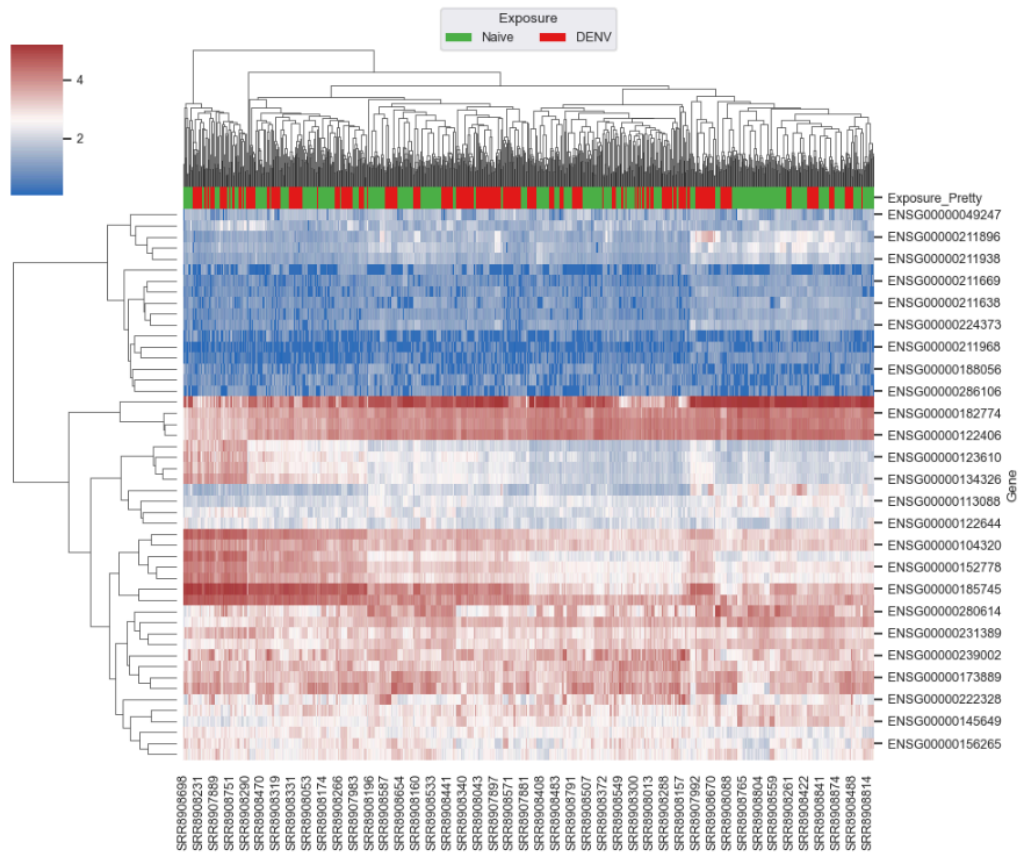
This plot illustrates how model AUC varies with the number of top variable genes used as features.

Performance improved sharply up to around 1000 genes, after which the AUC slightly declined, showing diminishing returns with excessive features.

The results suggest that 100–1000 top genes provide the best balance between signal and noise.

Overall, this experiment highlights the importance of dimensionality reduction in improving model efficiency and interpretability.

Top Predictive Genes Heatmap with Sample Clustering



The heatmap visualizes the top 50 most predictive genes identified by the Random Forest model, clustered across all samples.

You can see distinct expression patterns are visible between the **Naive** (green) and **DENV** (red) exposure groups, with samples naturally grouping by condition.

Both the row and column dendrograms reveal strong structure, suggesting consistent co-expression among key predictive genes.

This clustering supports the model's high classification accuracy, showing that a small subset of genes captures a surprisingly meaningful separation between the two exposure states.